Decision Making at a glance

- Branching
  - If/else
    - slides 30 - 32
  - switch
  - goto
- Looping
  - while
    - slide 45
  - do/while
  - for
- Jumps in loops
  - break

Do-while loop

- Do-while is an exit controlled loop
  - (while is entry controlled)
  - When loop content should execute at least once
- E.g
  - Looping until user enters valid value
  - Gather game input for game loop

```
do
{
    //loop body
} while (condition);
```

Do while : example

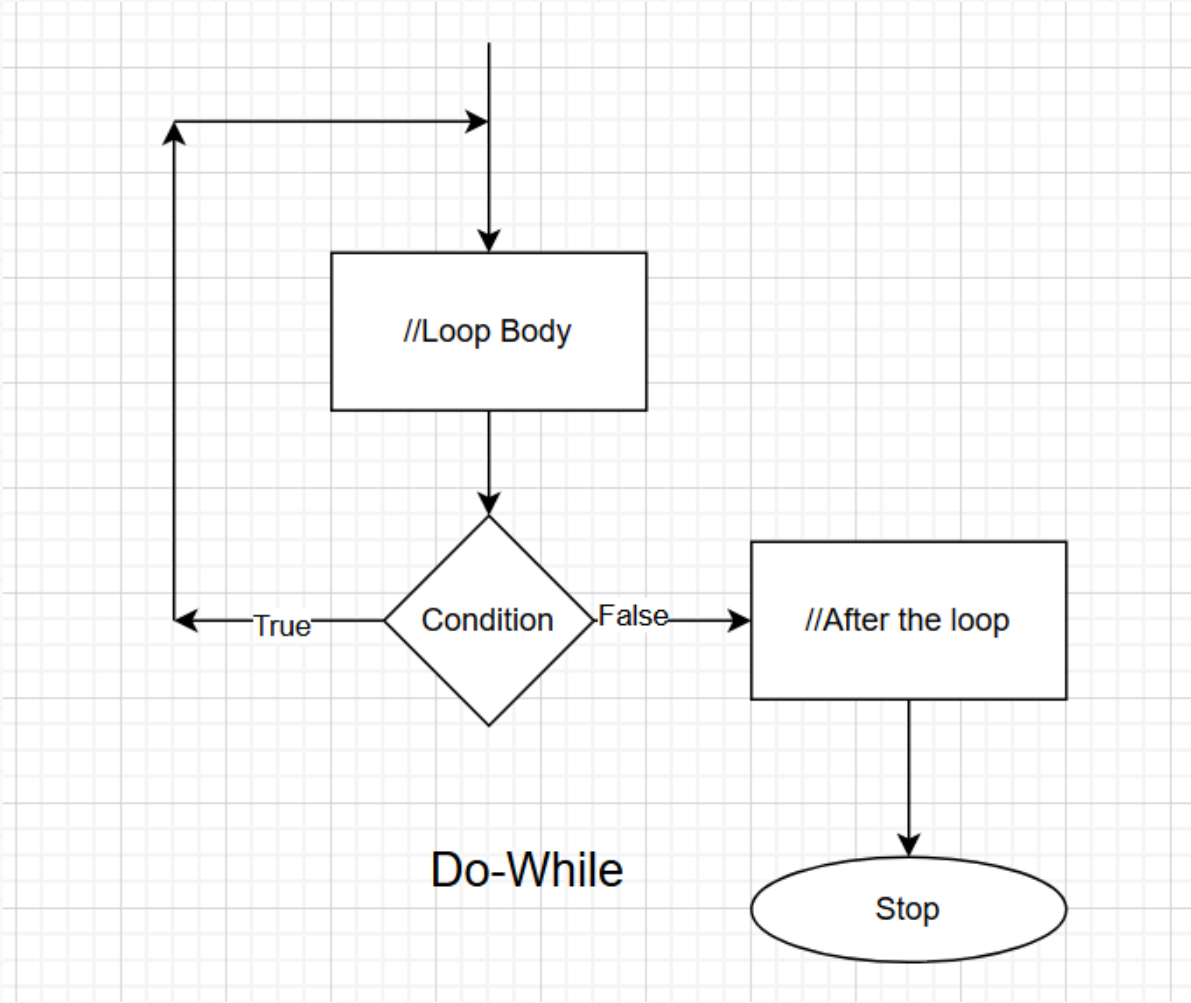
- Get a number between 1 and 4 from user.
  - Ask again if the numbers is not in range
  - Print the number and stop if it is in range.

```
int num = 0;

do
{
    cout << "\n Enter a number between 1 and 4 :";
    cin >> num;
} while ( num<1 || num >4);

cout << "\n You entered : " << num;
```

While vs do-while, visualized



Jumps in loops

- Break
  - Jump to after the end of loop
  - Loop does not execute
- Continue
  - Jump to the beginning of the loop
  - Loop continues to execute

```
int num;
while (true) { // An infinite loop
    cout << "Enter a number (0 to stop): ";
    cin >> num;
    if (num == 0) {
        break; // Immediately exit the loop when the condition is met
    }
    //Process loop here
}
```

```
do
{
    cout << "\n this is even number detector";
    cin >> num;

    if (num % 2 != 0)
    {
        cout << "\n this is not an even number";
        continue; //Continue executing the loop
    }
    cout << "\n this is an even number";
```

PRACTICE

Assignment 1: Create a menu driven calculator

- Enter 2 numbers
- Get user option (0 to 5) repeatedly
  - 1 -> add the two numbers and print
  - 2 -> subtract 2nd from first number and print
  - 3 -> multiply the two numbers and print
  - 4 ->divide 2nd by first number and print
  - 5 ->do nothing , and restart
  - 0 ->stop the loop
- Try using both loops and compare result
  - while
  - Do/while
- Use break and continue

Boss Assignment: Menu driven multi purpose program

- Create a multi purpose program running until user enters 0
- Enter 2 numbers e.g. a, b;
- Get user options repeatedly and perform the operation
  - 1 -> find the least common multiple of 2 numbers
  - 2 -> find the highest common factor of the 2 numbers
  - 3 -> find ‘a' raised to the power of ‘b' using loops : ab
  - 4 -> find 1/ab using loops
  - 5 -> do something random from between 1 and 4
    - #include<cstdlib >
    - int rand = 1+ rand()%4;
  - 0 -> Exit the program